C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 1.01

Functions, Modulus & Quadratics

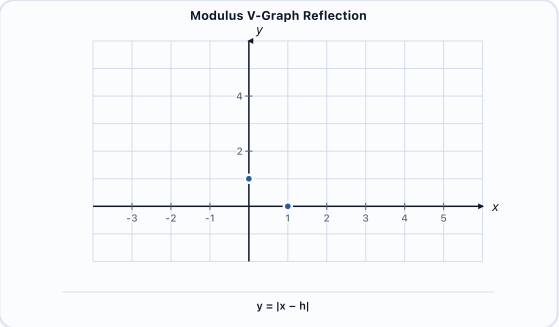
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Functions, Modulus & Quadratics Practice Exercises

1 MCQ

The function $f(x) = \frac{3x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



A $f^{-1}(x) = \frac{3x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-3}$

C $f^{-1}(x) = \frac{x-2}{3x+1}$

D $f^{-1}(x) = \frac{2x+1}{x-3} \quad (x \neq 3)$

2 MCQ

The function $f(x) = \frac{4x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x-1}{x-4}$

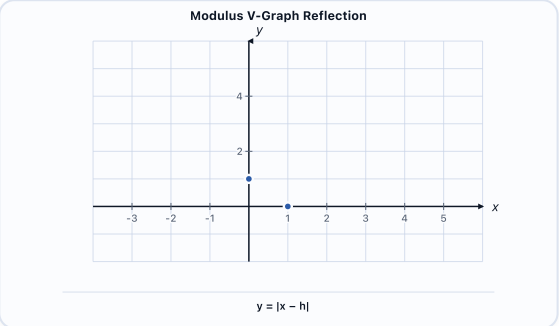
B $f^{-1}(x) = \frac{x-2}{4x+1}$

C $f^{-1}(x) = \frac{2x+1}{x-4} \quad (x \neq 4)$

D $f^{-1}(x) = \frac{4x-2}{x+1}$

3 MCQ

The function $f(x) = \frac{5x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



A $f^{-1}(x) = \frac{2x+1}{x-5} \quad (x \neq 5)$

B $f^{-1}(x) = \frac{x-2}{5x+1}$

C $f^{-1}(x) = \frac{5x-2}{x+1}$

D $f^{-1}(x) = \frac{2x-1}{x-5}$

4 MCQ

The function $f(x) = \frac{6x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-6} \quad (x \neq 6)$

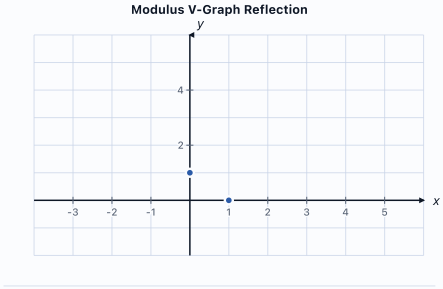
B $f^{-1}(x) = \frac{x-2}{6x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-6}$

D $f^{-1}(x) = \frac{6x-2}{x+1}$

5 MCQ

The function $f(x) = \frac{7x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.



A $f^{-1}(x) = \frac{7x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-7}$

C $f^{-1}(x) = \frac{x-2}{7x+1}$

D $f^{-1}(x) = \frac{2x+1}{x-7} \quad (x \neq 7)$

6 MCQ

The function $f(x) = \frac{8x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-8} \quad (x \neq 8)$

B $f^{-1}(x) = \frac{x-2}{8x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-8}$

D $f^{-1}(x) = \frac{8x-2}{x+1}$

7 MCQ

The function $f(x) = \frac{9x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x-1}{x-9}$

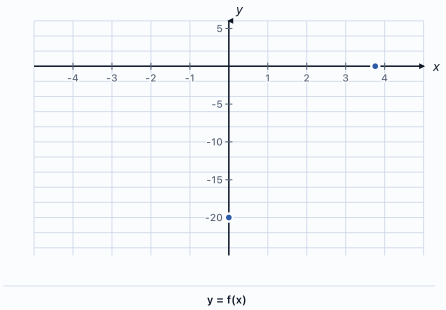
B $f^{-1}(x) = \frac{2x+1}{x-9} \quad (x \neq 9)$

C $f^{-1}(x) = \frac{x-2}{9x+1}$

D $f^{-1}(x) = \frac{9x-2}{x+1}$

8 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x - 2)$.



A -10

B -7

C -12

D -22

9 MCQ

The quadratic graph $f(x) = 4x^2 + gx + 256$ is tangent to the horizontal axis with $g > 0$. Find the positive coefficient g.

A 64

B 68

C 72

D 54

10 MCQ

Find the set of values of the constant k for which the equation

$$|x^2 - 4| = 2x + k$$

has exactly four distinct real solutions.

A $4 \leq k \leq 5$

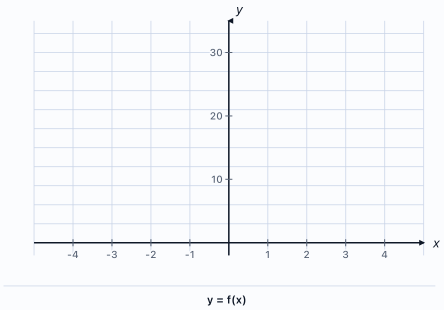
B $4 < k < 5$

C $-4 < k < 4$

D $k > 5$

11 MCQ

In the plane, a parabola is given by $f(x) = 3(x-7)^2 + 17$. Find the vertical intercept $f(0)$.



A 171

B 147

C 164

D 167

12 MCQ

Find the set of values of the constant m for which the equation

$$|2x-6| = mx+2$$

has exactly two distinct real solutions.

A $m < -\frac{2}{3}$ or $m > 2$

B $-\frac{2}{3} \leq m \leq 2$

C $-2 < m < 2$

D $-\frac{2}{3} < m < 2$

13 MCQ

The function f is defined for $x \in \mathbb{R}$ by

$$f(x) = x^2 - 2kx + (3k+10),$$

where k is a real constant, and the equation $f(x) = 0$ has no real roots.
The function g is defined for $x \in \mathbb{R}$ by

$$g(x) = |x-k|.$$

Find the set of values of k for which the minimum value of the composite function $h(x) = f(g(x))$ is strictly greater than 4.

A $\frac{3-\sqrt{33}}{2} < k < \frac{3+\sqrt{33}}{2}$

B $-2 < k < \frac{3+\sqrt{33}}{2}$

C $-2 < k < 5$

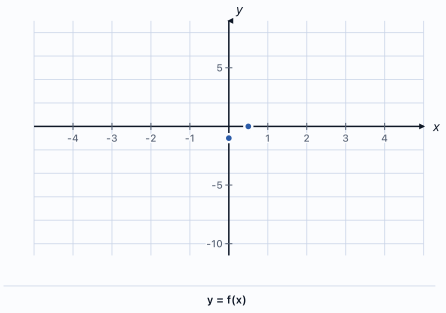
D $0 \leq k < \frac{3+\sqrt{33}}{2}$

14 MCQ

The functions f and g are defined for all real values of x by

$$f(x) = 2x - 1 \quad \text{and} \quad g(x) = \frac{x+5}{3}.$$

Find the value of x such that $fg(x) = 9$.



A $x = 9$

B $x = 5$

C $x = \frac{23}{2}$

D $x = 10$

15 MCQ

The quadratic equation $x^2 - (2k+1)x + (k^2+2) = 0$ has real roots α and β. Given that $\alpha^2 + \beta^2 = 27$, find the value of the constant k.

A 3

B -5

C 3 or -5

D 5

16 MCQ

In this problem, the function $f(x) = \frac{4x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x-1}{x-4}$

B $f^{-1}(x) = \frac{2x+1}{x-4} \quad (x \neq 4)$

C $f^{-1}(x) = \frac{x-2}{4x+1}$

D $f^{-1}(x) = \frac{4x-2}{x+1}$

17 MCQ

Observe that the function $f(x) = \frac{5x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{x-2}{5x+1}$

B $f^{-1}(x) = \frac{2x+1}{x-5} \quad (x \neq 5)$

C $f^{-1}(x) = \frac{5x-2}{x+1}$

D $f^{-1}(x) = \frac{2x-1}{x-5}$

18 MCQ

Suppose the function $f(x) = \frac{6x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{x-2}{6x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-6}$

C $f^{-1}(x) = \frac{2x+1}{x-6} \quad (x \neq 6)$

D $f^{-1}(x) = \frac{6x-2}{x+1}$

19 MCQ

Notice that the function $f(x) = \frac{7x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{7x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-7}$

C $f^{-1}(x) = \frac{x-2}{7x+1}$

D $f^{-1}(x) = \frac{2x+1}{x-7} \quad (x \neq 7)$

20 MCQ

Consider the function $f(x) = \frac{8x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{2x+1}{x-8} \quad (x \neq 8)$

B $f^{-1}(x) = \frac{x-2}{8x+1}$

C $f^{-1}(x) = \frac{2x-1}{x-8}$

D $f^{-1}(x) = \frac{8x-2}{x+1}$

21 MCQ

In this problem, the function $f(x) = \frac{9x+1}{x-2}$ is defined for $x \neq 2$. Find the inverse function $f^{-1}(x)$.

A $f^{-1}(x) = \frac{9x-2}{x+1}$

B $f^{-1}(x) = \frac{2x-1}{x-9}$

C $f^{-1}(x) = \frac{2x+1}{x-9} \quad (x \neq 9)$

D $f^{-1}(x) = \frac{x-2}{9x+1}$

22 MCQ

Solve the modulus equation $|2x + 6| = 4$.

A $x = -5$ or $x = -1$

B $x = -3$ or $x = 1$

C $x = 5$ or $x = 1$

D $x = -1$ only

23 MCQ

Solve the modulus equation $|3x + 6| = 6$.

A $x = -2$ or $x = 2$

B $x = -4$ or $x = 0$

C $x = 4$ or $x = 0$

D $x = 0$ only

24 MCQ

Solve the modulus equation $|4x + 4| = 8$.

A $x = -3$ or $x = 1$

B $x = 3$ or $x = -1$

C $x = -1$ or $x = 3$

D $x = 1$ only

25 MCQ

Solve the modulus equation $|5x| = 10$.

A $x = 2$ or $x = -2$

B $x = 2$ only

C $x = 0$ or $x = 4$

D $x = -2$ or $x = 2$